Miloš Stanojević

Associate Professor at UCL
Senior Research Scientist at Google DeepMind

Research Interests

- Linguistics: Language Universals, Syntactic Formalisms and Incremental Parsing
- Learning: Neural Latent Variable Models and Probabilistic Inference over Structures
- Applications: Machine Translation, Sign Language Generation and LLMs in general

Work Experience

Sep 2024 – now	**Associate Professor in Computational Linguistics** ***University College London*** Linguistics Department, London, UK
Nov 2021 – now	**Senior Research Scientist** ***Google DeepMind*** Language Team, London, UK
Nov 2017 – Oct 2021	**PostDoc / Research Associate** ***University of Edinburgh*** School of Informatics, Edinburgh, UK
July 2017 – Oct 2017	**Internship** ***Nuance Communications (now part of Microsoft)*** Natural Language and AI group, Sunnyvale, US

Education

2013 – 2017	**Ph.D. Natural Language Processing** ***University of Amsterdam, Netherlands***
2011 – 2013	**Mgr. Mathematical Linguistics** ***Charles University in Prague, Czechia***
2010 – 2011	**M.Sc. Human Language Science and Technology** ***University of Malta, Malta***
2006 – 2010	**B.Sc. Electrical Engineering** ***University of Niš, Serbia***

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PUBLICATIONS

EMNLP 2026 — **Syntactic Belief Update as the Driver of Garden Path Processing Difficulty**
EMNLP
Alan Zhou***, Miloš Stanojević***, John Hale
*equal contribution

Zenodo 2026 — **AISLAC Joint Impact Report for SL2T 1.0**
Zenodo
Google DeepMind Sign Language Team, DPAN, NAD, RIT/NTID, WFD

HSP 2026 — **Gone without a Trace: Why 5 LMs Fail to Track Non-Local Dependency Resolution**
Conference on Human Sentence Processing
Liliana Nentcheva, Donald Dunagan, **Miloš Stanojević**, Sebastian Schuster, Andrea Santi

HSP 2025 — **Generalizing active gap-filling**
Conference on Human Sentence Processing
Tim Hunter, **Miloš Stanojević**, Edward Stabler

COLING 2025 — **Extracting structure from an LLM - how to improve on surprisal-based models of Human Language Processing**
International Conference on Computational Linguistics
Daphne Wang, Mehrnoosh Sadrzadeh, **Miloš Stanojević**, Wing-Yee Chow, Richard Breheny

EMNLP 2024 — **Do LLMs learn a true syntactic universal?**
Empirical Methods in Natural Language Processing
John Hale and **Miloš Stanojević**

ACL 2024 — **Multipath parsing in the brain**
Annual Meeting of the Association for Computational Linguistics
Berta Franzluebbers, Donald Dunagan, **Miloš Stanojević**, Jan Buys, John Hale

CMCL 2024 — **How can large language models become more human?**
best paper award
Workshop on Cognitive Modeling and Computational Linguistics
Daphne Wang, Mehrnoosh Sadrzadeh, **Miloš Stanojević**, Wing-Yee Chow, Richard Breheny

EMNLP System Demo 2023 — **SynJax: Structured Probability Distributions for JAX**
Empirical Methods in Natural Language Processing System Demonstration
Miloš Stanojević and Laurent Sartran

Cognitive Science *journal* 2023 **top cited Cognitive Science paper in 2023**	**Modeling structure-building in the brain with CCG parsing and large language models** *Cognitive Science — journal* **Miloš Stanojević,** Jonathan Brennan**,** Donald Dunagan, Mark Steedman, John Hale
Neurobiology of Language *journal* 2023	**Neural correlates of long-distance dependencies in Chinese and English** *Neurobiology of Language — journal* Donald Dunagan, **Miloš Stanojević**, Maximin Coavoux, Shulin Zhang, Shohini Bhattasali, Jixing Li, Jonathan Brennan, John Hale
EMNLP 2022	**Unbiased and Efficient Sampling of Dependency Trees** *Conference on Empirical Methods in Natural Language Processing* **Miloš Stanojević**
TACL *journal* 2022	**Transformer Grammars: Augmenting Transformer Language Models with Syntactic Inductive Biases at Scale** *Transactions of the Association for Computational Linguistics — journal* Laurent Sartran, Samuel Barrett, Adhiguna Kuncoro, **Miloš Stanojević**, Phil Blunsom, Chris Dyer
Computational Linguistics *journal* 2021	**Formal Basis of a Language Universal** *Computational Linguistics — journal* **Miloš Stanojević** and Mark Steedman
EMNLP 2021	**A Root of a Problem: Optimizing Single-Root Dependency Parsing** *Conference on Empirical Methods in Natural Language Processing* **Miloš Stanojević** and Shay B. Cohen
IWCS 2021	**Computing All Quantifier Scopes with CCG** *International Conference on Computational Semantics* **Miloš Stanojević** and Mark Steedman
CMCL 2021	**Modeling incremental language comprehension in the brain with Combinatory Categorial Grammar** *Workshop on Cognitive Modeling and Computational Linguistics* **Miloš Stanojević**, Shohini Bhattasali, Donald Dunagan, Luca Campanelli, Mark Steedman, Jonathan Brennan, John Hale
CASE 2021	**Modality and Negation in Event Extraction** *Workshop on Challenges and Applications of Automated Extraction of Socio-political Events from Text* Sander Bijl de Vroe, Liane Guillou, **Miloš Stanojević**, Nick McKenna and Mark Steedman
ACL 2020	**Max-Margin Incremental CCG Parsing** *Annual Meeting of the Association for Computational Linguistics*

Miloš Stanojević and Mark Steedman

IWPT 2020 **Span-Based LCFRS-2 Parsing**
The 16th International Conference on Parsing Technologies
Miloš Stanojević and Mark Steedman

CUNY 2020 **Predictive Processing of Coordination in CCG**
Conference on Human Sentence Processing
Miloš Stanojević, John Hale and Mark Steedman

ACL 2019 **Wide-Coverage Neural A* Parsing for Minimalist Grammars**
Annual Meeting of the Association for Computational Linguistics
John Torr, **Miloš Stanojević**, Mark Steedman and Shay B. Cohen

FG 2019 **On the Computational Complexity of Head Movement and Affix Hopping**
Conference on Formal Grammar
Miloš Stanojević

NAACL 2019 **CCG Parsing Algorithm with Incremental Tree Rotation**
Conference of the North American Chapter of the Association for Computational Linguistics
Miloš Stanojević and Mark Steedman

CMCL 2019 **The active-filler strategy in a move-eager left-corner Minimalist Grammar parser**
Workshop on Cognitive Modeling and Computational Linguistics
Tim Hunter, **Miloš Stanojević** and Edward Stabler

CogACLL 2018 **A Sound and Complete Left-Corner Parsing for Minimalist Grammars**
Workshop on Cognitive Aspects of Computational Language Learning and Processing
Miloš Stanojević and Edward Stabler

PROPOR 2018 **SICK-BR: a Portuguese corpus for inference**
International Conference on Computational Processing of the Portuguese Language
Livy Real, Ana Rodrigues, Andressa Vieira e Silva, Beatriz Albiero, Bruna Thalenberg, Bruno Guide, Cindy Silva, Guilherme de Oliveira Lima, Igor CS Câmara, **Miloš Stanojević**, Rodrigo Souza, Valeria de Paiva

PhD Thesis 2017 **Permutation Forests for Modeling Word Order in Machine Translation**
University of Amsterdam 2017
Miloš Stanojević

EMNLP 2017 **Neural Discontinuous Constituency Parsing**

Conference on Empirical Methods in Natural Language Processing
Miloš Stanojević and Raquel G. Alhama

ACL 2017 short paper — **Alternative Objective Functions for Training MT Evaluation Metrics**
Annual Meeting of the Association for Computational Linguistics
Miloš Stanojević and Khalil Sima'an

LACL 2016 — **Minimalist Grammar Transition-Based Parsing**
Logical Aspects of Computational Linguistics
Miloš Stanojević

COLING 2016 — **Hierarchical Permutation Complexity for Word Order Evaluation**
International Conference on Computational Linguistics
Miloš Stanojević and Khalil Sima'an

COLING 2016 — **Universal Reordering via Linguistic Typology**
International Conference on Computational Linguistics
Joachim Daiber, **Miloš Stanojević** and Khalil Sima'an

WMT 2016 — **Examining the Relationship between Preordering and Word Order Freedom in Machine Translation**
Conference on Machine Translation
Joachim Daiber, **Miloš Stanojević**, Wilker Aziz and Khalil Sima'an

WMT 2016 — **Results of the WMT16 Metrics Shared Task**
Conference on Machine Translation
Ondřej Bojar, Yvette Graham, Amir Kamran and **Miloš Stanojević**

WMT 2016 — **Results of the WMT16 Tuning Shared Task**
Conference on Machine Translation
Bushra Jawaid, Amir Kamran, **Miloš Stanojević** and Ondřej Bojar

EMNLP 2015 — **Reordering Grammar Induction**
Conference on Empirical Methods in Natural Language Processing
Miloš Stanojević and Khalil Sima'an

PBML 2015 — **Evaluating MT systems with BEER**
The Prague Bulletin of Mathematical Linguistics
Miloš Stanojević and Khalil Sima'an

WMT 2015 — **BEER 1.1: ILLC UvA submission to metrics and tuning task**
Workshop on Machine Translation
Miloš Stanojević and Khalil Sima'an

WMT 2015 — **Results of the WMT15 Metrics Shared Task**
Workshop on Machine Translation
Miloš Stanojević, Amir Kamran, Philipp Koehn and Ondřej Bojar

WMT 2015	**Results of the WMT15 Tuning Shared Task** *Workshop on Machine Translation* **Miloš Stanojević**, Amir Kamran and Ondřej Bojar
SSST 2014	**Evaluating Word Order Recursively over Permutation-Forests** *Workshop on Syntax, Semantics and Structure in Statistical Translation* **Miloš Stanojević** and Khalil Sima'an
WMT 2014	**BEER: BEtter Evaluation as Ranking** *Workshop on Machine Translation* **Miloš Stanojević** and Khalil Sima'an
EMNLP 2014 short paper	**Fitting Sentence Level Translation Evaluation with Many Dense Features** *Conference on Empirical Methods in Natural Language Processing* **Miloš Stanojević** and Khalil Sima'an
WMT 2012	**Selecting data for English-to-Czech machine translation** *Workshop on Machine Translation* Aleš Tamchyna, Petra Galuščáková, Amir Kamran, **Miloš Stanojević**, Ondřej Bojar

Contributions to the Research Community

ARR Area Chair:	- ACL, NAACL, EMNLP, EACL, CoNLL conferences
Standing Reviewer:	- TACL journal (Impact Factor 6.9) - Computational Linguistics journal (Impact Factor 5.3)
Reviewer:	- Natural Language Processing Journal (NLP) (Impact Factor 1.9) - Language Resources and Evaluation Journal (LREC) (Impact Factor 1.8) - Journal of Language Modeling (Impact Factor not available)
Shared tasks organization:	- Workshop on Machine Translation (WMT) - Metrics task in 2015 & 2016 - Workshop on Machine Translation (WMT) - Tuning task in 2015 & 2016

Supervision and Mentorship

Johns Hopkins University	2026- **Alan Zhou (part of PhD dissertation committee)**
UCL supervision	2026- **Francesco Cipriani (PhD 2nd co-supervisor)**

	+ 8 MA students in Computational Linguistics specialization + 2 BA student in General Linguistics
UCL mentorship	2026 **Tal Nahari** now Postdoctoral Fellow, Affective Brain Lab, UCL
	2025 **Maximilian Moser** now Research at King's College London
Google DeepMind internship mentor of PhD students	2026 **Yifu Qiu** now PhD at University of Edinburgh
	2025 **Zifan Jiang** now Research Scientist at Google DeepMind
	2024 **Matthias Lindemann** now PostDoc at Instituto de Telecomunicações, Lisbon, Portugal
Google DeepMind scholar mentor	2026 **Sindhu B. Hegde** now PhD student at University of Oxford
	2023 **Catherine Nana Nyaah Essuman** now PhD student at Ontario Tech University
	2022 **Mukhtar Mohamed** now intern at Google DeepMind
University of Edinburgh supervision	2021 **Seth Aycock** (MSc Dissertation main supervisor) now PhD student at University of Amsterdam
	2019 **Chiehmin Wei** (BA project supervisor) now software engineer at Roblox

TEACHING

2025/2026 UCL	Lectures on Language Modeling within **PLIN0072 Advanced Computational Linguistics**
2025/2026 UCL	Lectures on Parsing within **PLIN0034 Introduction to Computational Linguistics**
2024/2025 *UCL*	Lectures on Parsing within **PLIN0034 Introduction to Computational Linguistics**
2024/2025 UCL	Present linguistics students with career opportunities in the industry **UCL LingSoc Careers Event**

2016/2017 *University of Amsterdam*	**Natural Language Processing 2**
2015/2016 *University of Amsterdam*	**Natural Language Processing 2**
2014/2015 *University of Amsterdam*	short course on **Monte Carlo Inference for PCFG**
2013/2014 *University of Amsterdam*	**Elements of Language Processing**
2012/2013 *University of Amsterdam*	**Statistical Structure in NLP**

Invited Talks

LSSC keynote speaker *Novi Sad, Serbia, 2026*	**Interaction of Linguistics and Computer Science Then and Now** Keynote speaker at conference *Language & Social Sciences in Contact: Artificial Intelligence (AI)*
FlaNN Seminar *Online, 2026*	**Modeling Sentence Processing Difficulty with Bayesian Syntax** *Formal Languages and Neural Networks Seminar*
Stony Brook University *New York, USA, 2025*	**Incremental Sentence Processing of Right-Adjunction**
ILFC Seminar – CNRS, *Paris, France, 2025*	**Linguistic Universals in Grammars and Language Models** *ILFC Seminar: Interactions between Formal and Computational Linguistics*
Johns Hopkins University *Baltimore, USA, 2024*	**Syntactic Universals in Grammars and LLMs**
University of Amsterdam *Netherlands, 2023*	**Syntactic Structures and Large Language Models**
University of Cambridge *Cambridge, UK, 2023*	**The role of syntax in the world of large language models**
UCL *London, UK, 2023*	**What is the relation of Syntactic Parsing, Language Universals, Brain Activity and Large Language Models?**
ETH Zürich Zürich, Switzerland, 2022	**Testing Syntactic Theory on all Marr's Levels of Analysis**
Mathematics of Language Montpellier, France, 2021	**CCG Combinators in Brain Activity and Linguistic Universals**

DeepMind London, UK, 2021	**Integrating Deep Learning with Deep Syntax**
Nuance NL & AI lab Sunnyvale, USA, 2017	**Learning Reordering Grammars**
Rich Parsing Workshop Amsterdam, 2016	**Minimalist Grammar Transition-Based Parsing**
MT Marathon Prague, Czechia, 2015	**Discriminative Training for Statistical Machine Translation**

GRANTS

2026	**Capital Equipment Investment Fund by UKRI** £185,025 for computing equipment Miloš Stanojević (lead applicant) and Mario Giulianelli (co-applicant)
2025	**Language Sciences Incubator Fund by University of Cambridge** £5,000 for research project "Improving Language Equity and Inclusion through AI" lead applicants: Songqiao Xie and Xinxin Yan co-applicants: Mario Giulianelli, Napoleon Katsos, Miloš Stanojević

SCHOLARSHIPS

2010 – 2012	**Erasmus Mundus scholarship for EM-LCT Masters Program**
2004 – 2010	**Scholarship from the Ministry of Science and Technological Development of Serbia** awarded for taking part in international mathematical competitions
2002 – 2004	**Scholarship from the Ministry of Education of Serbia** awarded for taking part in international mathematical competition